Parametric Equations

ANSWER KEY

Section 1: First and Second Derivatives Parametrically

1 A curve is defined by $x(t) = t^2 + 1$ and $y(t) = t^3 - 3t$. Find $\frac{dy}{dx}$ at $t = 2$.

a. $\frac{9}{4}$ ✓

b. $\frac{3}{2}$

c. $\frac{4}{9}$

d. 9

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 3t^2 - 3$. $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2 - 3}{2t}$. At $t = 2$: $\frac{12 - 3}{4} = \frac{9}{4}$.

2 Which formula gives the second derivative $\frac{d^2y}{dx^2}$ in parametric form?

a. $\frac{\frac{d}{dt}[\frac{dy}{dx}]}{\frac{dx}{dt}}$ ✓

b. $\frac{d^2y/dt^2}{d^2x/dt^2}$

c. $\frac{d}{dt}[\frac{dy}{dx}]$

d. $\frac{\frac{dy}{dt}}{\frac{d^2x}{dt^2}}$

By the chain rule, $\frac{d^2y}{dx^2} = \frac{d}{dx}[\frac{dy}{dx}] = \frac{\frac{d}{dt}[\frac{dy}{dx}]}{\frac{dx}{dt}}$.

3 For $x(t) = t^2$ and $y(t) = t^3$, find $\frac{d^2y}{dx^2}$.

a. $\frac{3}{4t}$ ✓

b. $\frac{3}{2}$

c. $\frac{6t}{2}$

d. $\frac{3}{4t^2}$

$\frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3}{2}t$. Then $\frac{d}{dt}[\frac{3}{2}t] = \frac{3}{2}$. Dividing by $\frac{dx}{dt} = 2t$ gives $\frac{3/2}{2t} = \frac{3}{4t}$.

Section 2: Horizontal and Vertical Tangents

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 At what value of t does the parametric curve $x = t^3 - 3t, y = 2t^2$ have a horizontal tangent line?

a. $t = 0$ ✓

b. $t = 1$

c. $t = -1$

d. $t = \pm 1$

A horizontal tangent requires $\frac{dy}{dt} = 0$ while $\frac{dx}{dt} \neq 0$. $\frac{dy}{dt} = 4t = 0 \Rightarrow t = 0$. Here $\frac{dx}{dt}(0) = -3 \neq 0$.

5 At what values of t does the curve above have vertical tangent lines?

a. $t = \pm 1$ ✓

b. $t = 0$

c. $t = 3$

d. $t = \pm\sqrt{3}$

Vertical tangents occur where $\frac{dx}{dt} = 0$ while $\frac{dy}{dt} \neq 0$. $\frac{dx}{dt} = 3t^2 - 3 = 0 \Rightarrow t = \pm 1$. $\frac{dy}{dt}(\pm 1) = \pm 4 \neq 0$.

6 Find the equation of the tangent line at $t = 1$ to the curve $x(t) = e^{2t}, y(t) = 3t^2 - 1$.

a. $y - 2 = \frac{3}{e^2}(x - e^2)$ ✓

b. $y - 2 = 3(x - e^2)$

c. $y = \frac{6}{e^2}x$

d. $y - 1 = \frac{3}{e^2}(x - e^2)$

At $t = 1$: $x = e^2, y = 2$. $\frac{dx}{dt} = 2e^{2t} = 2e^2$, $\frac{dy}{dt} = 6t = 6$. Slope $= \frac{6}{2e^2} = \frac{3}{e^2}$. Equation: $y - 2 = \frac{3}{e^2}(x - e^2)$.

Section 3: Arc Length of Parametric Curves

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Which integral represents the arc length L of a parametric curve from $t = a$ to $t = b$?

a. $\int_a^b \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2}\,dt$ ✓

b. $\int_a^b (\frac{dx}{dt} + \frac{dy}{dt})dt$

c. $\int_a^b \sqrt{1 + (\frac{dy}{dx})^2}\,dt$

d. $\int_a^b \sqrt{x(t)^2 + y(t)^2}\,dt$

By the distance formula differential $ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{(x'(t))^2 + (y'(t))^2}\,dt$, the arc length is $\int_a^b \sqrt{(x')^2 + (y')^2}\,dt$.

8 Find the arc length of the curve $x = 3\cos t, y = 3\sin t$ from $t = 0$ to $t = \pi$.

a. 3π ✓

b. 6π

c. 9π

d. $\frac{3\pi}{2}$

$x' = -3\sin t, y' = 3\cos t$. $\sqrt{(-3\sin t)^2 + (3\cos t)^2} = \sqrt{9} = 3$. Length $= \int_0^\pi 3dt = 3\pi$ (semicircle of radius 3).

9 Find the arc length of $x = 2t^{3/2}, y = 3t$ from $t = 0$ to $t = 3$.

a. 14 ✓

b. 16

c. 18

d. 12

$x' = 3\sqrt{t}$, $y' = 3$. Speed $= \sqrt{9t + 9} = 3\sqrt{t + 1}$.
$L = \int_0^3 3\sqrt{t+1}\,dt = [2(t+1)^{3/2}]_0^3 = 2(4^{3/2} - 1^{3/2}) = 2(8 - 1) = 14$.

Section 4: Speed and Total Distance Traveled

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 A particle moves in the xy-plane with velocity vector $\mathbf{v}(t) = \langle 4t, 3 \rangle$. What is its speed at $t = 1$?

a. 5 ✓

b. 7

c. 4

d. $\sqrt{7}$

Speed is the magnitude of the velocity vector: $\|\mathbf{v}(1)\| = \sqrt{(4(1))^2 + 3^2} = \sqrt{16 + 9} = 5$.

11 What is the total distance traveled by the particle with velocity $\mathbf{v}(t) = \langle 2t, 2 \rangle$ from $t = 0$ to $t = 2$?

a. $\int_0^2 \sqrt{4t^2 + 4}\, dt$ ✓

b. $\int_0^2 (2t + 2) dt$

c. $\sqrt{16 + 4}$

d. $2\sqrt{5}$

Total distance traveled is the integral of speed: $\int_0^2 \sqrt{(2t)^2 + 2^2}\, dt = \int_0^2 \sqrt{4t^2 + 4}\, dt$.

12 When is a parametric particle at rest in the coordinate plane?

a. **When both $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$ simultaneously** ✓

b. When $\frac{dy}{dx} = 0$

c. When $\frac{dx}{dt} = 0$

d. When speed is undefined

A particle is at rest if and only if both component velocities are simultaneously zero, meaning speed $\|\mathbf{v}(t)\| = 0$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647